

assignments in

Junior Science

book 1
Electricity

SUADA BILALI
CHRIS GREEF



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**SUADA BILALI
CHRIS GREEF**



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Introduction



Each book in the *Assignments in Junior Science* series provides a selection of worksheets covering one or two major topics. The worksheets are independent, supplying most information necessary to formulate answers. In the answer section at the back of each book, sample answers to most tasks have been given.

Many worksheets have an emphasis on literacy, others concentrate on problem solving, while others utilise science in an experiment or a game format. They are presented in a student-friendly manner and could be used to supplement science lessons for regular class teachers or casual teachers.

Suada Bilali and Chris Greef are both experienced and practising science teachers who had a lot of fun putting together this collection of worksheets.

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- 3 Summarise the contribution of each scientist in a table. Include some detail about the scientists.

Name	Date	Country	Contribution

- 4 Identify the scientist who lived the longest and the scientist who had the shortest life. State their ages.

Looking at the language

- 5 Underline the verb groups in paragraph 1. Three examples have been done for you already.
- 6 Select the correct verb form from each selection in paragraph 2 and underline it. One example has been done for you. Then rewrite the passage using only the correct verbs.
- 7 There are four main types of verbs:
- action verbs describe events;
 - linking verbs join characteristics to things;
 - saying verbs describe how something is said;
 - thinking verbs show feelings and thoughts.
- The text above is an historical recount, which usually has all four types of verbs. Complete the table below by identifying three examples for each verb type. One example is already given.

Action verb	Linking verb	Saying verb	Thinking verb
build	is made up of	shout	feel

- 8 In an *active voice*, the person performing the action is placed at the front of the sentence. In *passive voice* that person is placed at the back of the sentence, or left out altogether.

Examples: The scientist measured the electric current. [ACTIVE VOICE]

The electric current was measured (by the scientist). [PASSIVE VOICE]

Rewrite the first and last sentences of paragraph 3 of the text, changing the voice.

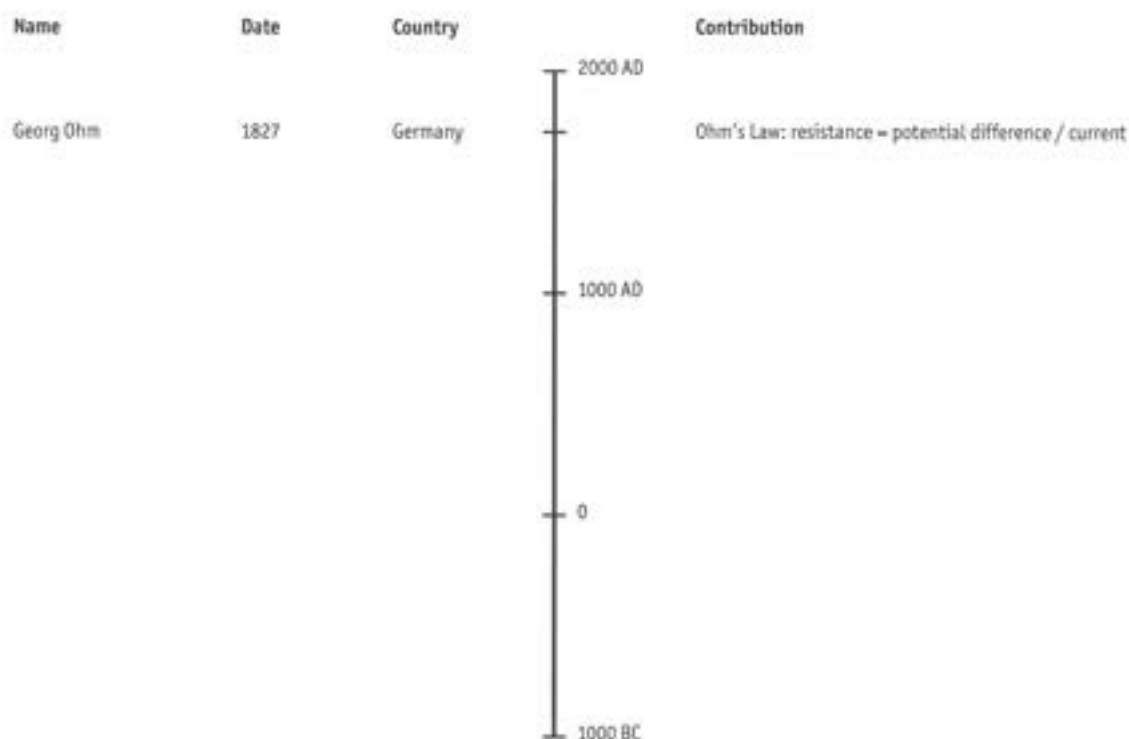
Whose bright idea?

2



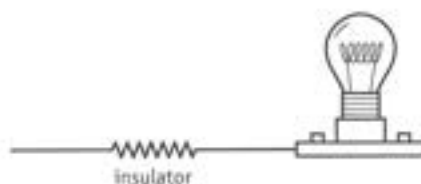
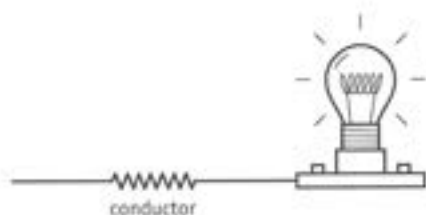
Alessandro Volta (Italy, 1745 – 1827) separated two metal discs with a cloth that was moistened with saltwater, and measured very weak electricity between the discs. However, in 1796, when he piled up a battery of paired silver and zinc discs, he not only found that the electricity was stronger, but also that the electricity flowed for several minutes. This 'voltaic pile' was the first electric battery.	In 1820 the Danish scientist Hans Christian Oersted (1777 – 1851) noticed that a magnetic compass needle turned when he connected a voltaic pile battery with a wire. He was sure that electricity and magnetism were connected and that this connection had to do with the movement of electric charge.
In 1820 Andre Ampere (France, 1775 – 1836) found that the interaction between electricity and magnetism was strengthened when the wire was made into a loop. A loop of wire carrying electricity acted just like a magnet.	In 1825 the English scientist William Sturgeon (1783–1850) wound a coil of wire around an iron bar, which made a very strong magnet when electricity flowed. In fact, Sturgeon had built the first electromagnet.
The idea of an electromagnet was also used by Michael Faraday (England, 1791 – 1867), when he built simple electric motors and dynamos in 1828.	The idea of an electromagnet was used by Joseph Henry (New York, 1797 – 1878) in 1828, when he wound a huge coil of wire around an iron bar and then lifted 1 ton of iron.
In 1752 American scientist Benjamin Franklin (1706 – 1790) showed that thunderstorms were electrical.	In 1786 the Italian anatomist Luigi Galvani (1737 – 1798) experimented with frog legs that were freshly cut and found that an electric charge made the frog leg muscles twitch.
Georg Ohm (Germany, 1789 – 1854) discovered in 1827 that the resistance of an electrical appliance stayed the same and could be calculated by dividing the potential difference by the current. This is known as Ohm's Law.	The ancient Greeks knew about electricity: Thales of Miletus (625 – 550 BC) noticed in 530 BC that a piece of rubbed amber attracted tiny pieces of sawdust.

Refer to the snippets of information above and summarise them on the timeline below. Write the contributor's name, date and country of birth on the left side of the timeline and his contribution on the right side.



Our electrifying experiment

This morning Ms Smart wanted our science class to perform an experiment with electricity. After we collected all the equipment, I set up the electrical circuit for the experiment. I put one electrical wire in the positive terminal of the powerpack (Ms Smart said to use the red and black DC terminals) and then connected it to a light bulb. Another wire, which started at the lamp, had an alligator clip on its other end. The third wire, which also had an alligator clip on its end, came off the negative terminal of the powerpack. My friend then had to check that the electrical circuit was switched correctly. It was all correct. We then clipped a piece of copper between the two alligator clips, turned on the power and saw that the lamp lit up brightly. We turned the power off again and put a piece of wood between the alligator clips. This time the lamp didn't come on at all when we turned the power on. After that we tried some aluminium foil, which worked as well as copper. But using a piece of plastic our bulb did not light up. We tried lots of other materials: carbon worked, cardboard did not, iron worked again, but not rubber, cotton or some thick string. After we had recorded our findings in a results table, we put all the equipment away again. Finally, Ms Smart drew two diagrams (which I have copied below) on the board with one word below each. She said we were to use these definitions and then add two more columns to our table. They were the conclusions for our experiment. We used ticks for 'yes' and crosses for 'no' in the table.



Recording the experiment scientifically

- Experiments are usually written up using the text type 'experimental record'. An experimental record has the following subheadings: Aim, Conclusion, Diagram, Equipment, Method, Results. Write the appropriate subheadings in the boxes above questions 2 to 7 below.

- For what purpose did the class do the experiment? (Complete the sentence.)

The aim was _____

- What equipment did the students use?

Apparatus: _____

Materials: _____

- What did the students have to do? Record the method of the experiment as if you are giving orders to the person performing the experiment.



- 5** What did the students' experiment look like? Sketch a labelled diagram.



- 6** What observations did the students make? Write two sentences to summarise their results.



- 7** What was the conclusion of the experiment? Write one sentence.

- 8** Scientists often record answers to repetitive tasks in tables. Complete the table below to show the observations and conclusions.

Materials	Observations	Conclusions

Definitions

- 9** What is a conductor? Define the term.

- 10** What is an insulator? Define this term.

Applications

- 11** Electrical wires carry electricity from one place to another. The electrical wires in the experiment were made of thin copper wires with some coloured plastic around them. The wires would have been much cheaper to make if they didn't have the plastic around them, so there must be a good reason for the plastic being there. Suggest a reason.

- 12** Computer chips contain tiny microscopic electrical circuits. Some chips use gold instead of copper for the wires. Since gold is much more expensive than copper, gold must have an advantage over copper. Suggest such an advantage.

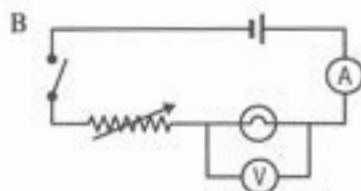
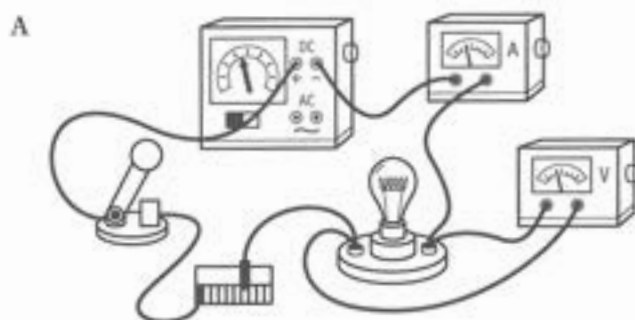
Circuit symbols

4a

An easy way to show how electrical devices are connected to a power source is by drawing a circuit diagram. A circuit diagram is made up of circuit symbols. These symbols are common and accepted by people from countries all over the world. The components or parts that make up a circuit can be classified into four groups:

1. **Energy suppliers:** these supply the electrical energy to make the circuit work.
2. **Measuring devices:** these components make measurements in a circuit.
3. **Energy users:** these devices change electrical energy into other forms of energy, such as heat or light.
4. **Circuit controllers:** these can turn off the current or control the amount of energy in a circuit.

1 Carefully observe the two diagrams A and B below.



- a What is similar about them? _____
- b What is different about them? _____

2 Use diagrams A and B above to work out the symbols of the electrical devices pictured. Draw the symbols in the space provided in the table.

Part	Use	Symbol	Group
Ammeter	Measures electrical current		
Battery or powerpack	Produces electrical energy		
Globe	Changes electrical energy into light and heat		
Switch	Is a gate in the circuit used by people to turn off the current		
Variable resistor	Controls the amount of energy in the circuit		
Voltmeter	Measures electrical potential difference		

3 Into which of the four circuit groups could each device be classified? Add your answers to the last column of the table.

Circuit symbols

4b



4 Why do scientists or technicians prefer to use the style of diagram B?

5 Draw a circuit diagram for the following set-ups.

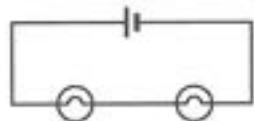
Circuit	Circuit diagram

Series and parallel circuits

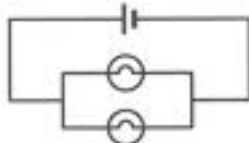
5a

A series circuit means that all components are connected end to end in the same circle, that is there is only one pathway. In a parallel circuit, electrons are given a choice of an alternative route to get around the circuit; in other words they form two or more circles off the same power source. The other routes are called branches.

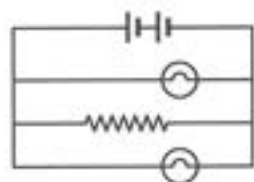
1 Label the diagrams below as either *in series* or *parallel* circuits.



a _____



b _____



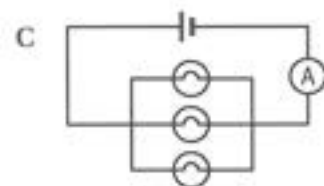
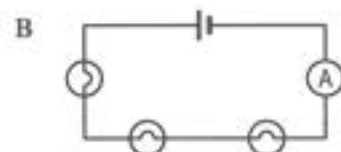
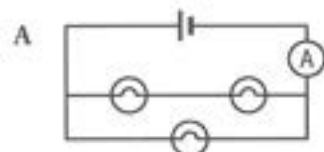
c _____



d _____

2 Match the descriptions to the diagrams by drawing lines between texts and diagrams.

- a This circuit is made up of a 12 volt battery that is powering three light globes in series. An ammeter is connected to measure the current going through the light globes.
- b This circuit is made up of a 12 volt battery that is powering three light globes in parallel. An ammeter has been connected to measure the total current flowing through the light globes.
- c This circuit is made up of a 12 volt battery powering three light globes, two of which are in series, with the third one in parallel. An ammeter is connected to measure the current entering the light globes.



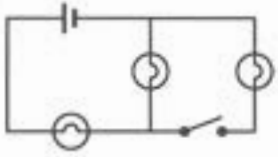
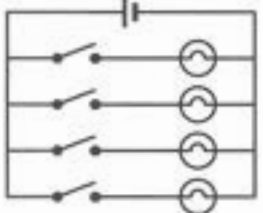
3 Draw the following circuit diagrams.

- a A 6 volt battery powering two lights and a variable resistor in parallel, with each light having its own switch.
- b A 12 volt battery powering two globes in series. The circuit also has an ammeter and a switch at one end of each globe.

Series and parallel circuits

5b

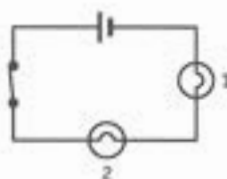
- 4 Write a description for each of these circuit diagrams.

Circuits	Descriptions
a 	
b 	

- 5 If one globe in a series circuit was to go out, the other globes would go out too. If one globe in a parallel circuit was to go out, the other globes in the circuit would stay lit. Why?

- 6 Observe the diagram to the right then answer the following questions.

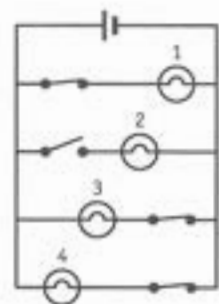
- a What kind of circuit is this? _____
- b How many possible paths can the electrons follow? _____
- c Electrons move from the negative terminal to the positive terminal. Draw arrows on the diagram to show the path of the current.
- d If globe 1 was removed, what would happen to globe 2? _____



- e If globe 1 was replaced with a wire, what would happen to the brightness of globe 2? _____

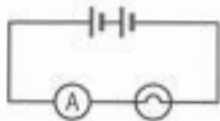
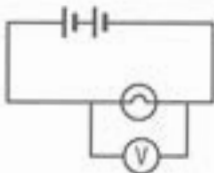
- 7 Observe the diagram to the right, then answer the questions.

- a What kind of circuit is this? _____
- b How many paths can the electrons follow? _____
- c If globe 2 was removed, what would happen to globes 1 and 3? _____
- d Looking at the diagram, which globe is not working? Why? _____
- e Describe the brightness of each globe. _____



- 8 Do you think your house would be wired like the diagram in question 6 or 7? Give a reason for your answer. _____



Ammeters	Voltmeters
 ■ Used to measure current (number of moving electrons) through an appliance. ■ Current is measured in amperes. (1 ampere means that 6 300 000 000 000 000 electrons pass by every second.) ■ Low resistance to electricity and should not be connected straight across the power supply. ■ Large amounts of current pass through. ■ Electric current is the same everywhere in a series circuit. ■ Energy is used up by the lamp. ■ Connected in series next to the appliance.	 ■ Used to measure potential difference (the amount of electrical potential or pressure between two points). ■ Potential difference is measured in volts. ■ High resistance to electricity and able to be connected straight across the power supply. ■ Tiny amounts of current pass through. ■ Potential difference varies for different appliances. ■ Energy is used up around the circuit. ■ Placed in parallel across the appliance.

1 The table provides summary notes about ammeters and voltmeters. Write the corresponding points into contrasting sentences using connectives such as: *but*, *and*, *on the other hand*, *while*. The first one has been done for you.

a Ammeters are used to measure current through an appliance, whereas voltmeters measure the difference in electrical potential between two points.

b _____

c _____

d _____

e _____

f _____

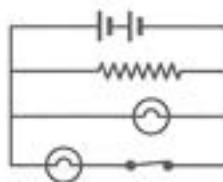
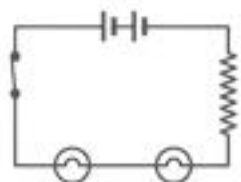
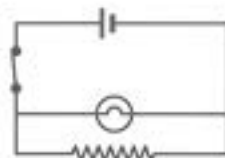
g _____

Ammeters and voltmeters

6b

2 For each of the diagrams add and draw:

- an ammeter to measure the current through the light globes and
- a voltmeter to measure the potential difference across the resistors.



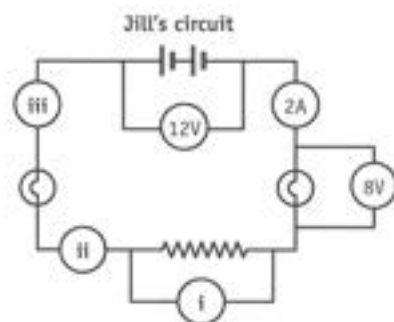
3 Jill set up the following circuit in Ms Smart's classroom, but only recorded three measurements and forgot to label some of the measuring devices. Look at the diagram and answer the questions.

a What sort of measuring devices are the following?

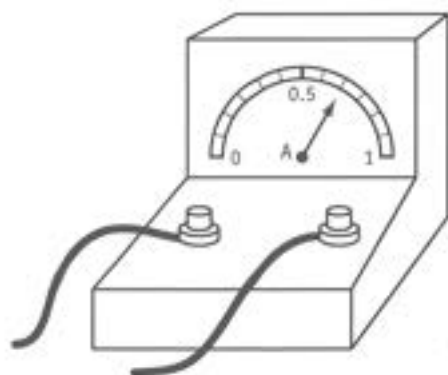
- _____
- _____
- _____

b What is the reading on the meters?

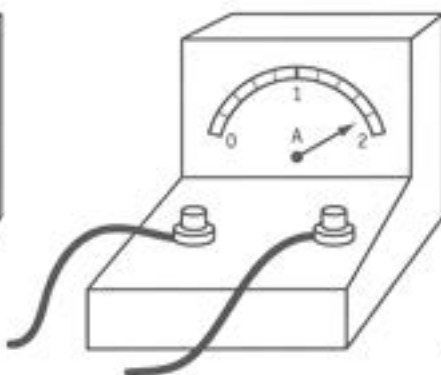
- _____
- _____
- _____



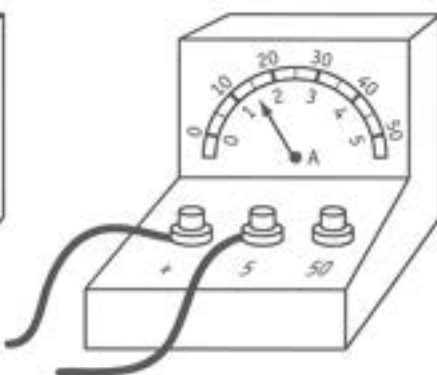
4 Write down the readings depicted on the following ammeters.



a _____



b _____



c _____

Circuits in text, sketches and diagrams 7a



1 Below is a table listing possible components of an electric circuit. Connect the boxes to link the appropriate name with its matching sketch, symbol and function.

	Name	Sketch	Symbol	Function
a	AC power source			Allows current to flow through.
b	ammeter			Changes electrical energy into light and heat energy.
c	DC power source			Interrupts flow of current when open.
d	fuse			Is a variable resistor slowing current to varying degrees.
e	light bulb			Measures current in amps and is switched in series.
f	resistor			Measures the potential difference in volts and is switched in parallel.
g	rheostat			Melts if too much current flows through it and as a result disrupts the circuit, preventing any further electricity to flow.
h	switch			Produces alternating current where current changes its direction continuously; can be a powerpack.
i	voltmeter			Produces direct current flowing in one direction only; can be a battery or powerpack.
j	wire			Slows current and changes some electrical energy.

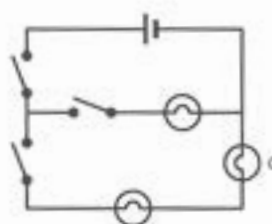
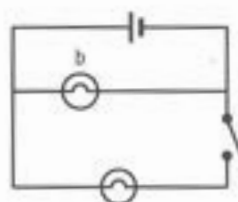
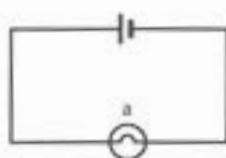
2 Complete the table below by writing some text to describe the circuit, by sketching the circuit as you would set it up in the laboratory, or by drawing an electrical circuit diagram.

Text	Sketch	Diagram
a An electric circuit is made up of three items connected by wires. The positive terminal of a powerpack is connected with a wire to a light bulb. The bulb is connected to a switch and the switch to the negative terminal.		
b		

Circuits in text, sketches and diagrams 7b

Text	Sketch	Diagram
c		
d This circuit has six items connected to a powerpack by wires. Two light bulbs are connected in series and then two more light bulbs are connected in parallel. There is one switch before each parallel bulb.		
e		
f		

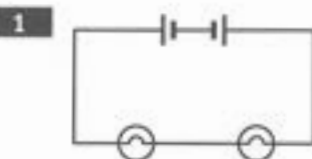
- 3** An ammeter measures current; in a way it is counting how many electrons are racing through it. Ammeters are switched in series with the device. The unit of current is amperes or amps (A). Voltmeters measure the potential difference; they indicate the tension or pressure between two points. Voltmeters are wired parallel with the device. Potential difference is measured in volts (V). Look at the diagrams below and draw in voltmeters and ammeters to take measurements of the devices numbered a, b and c.



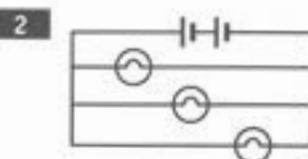
Changing circuits

8

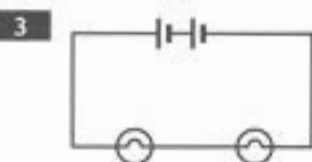
Look at the diagrams below. What would happen if the electric circuit was changed as indicated? Write your predictions in the spaces provided.



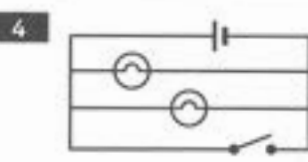
You connect the batteries this way.



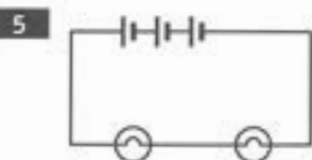
You add another globe.



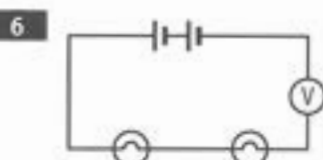
You loosen one globe.



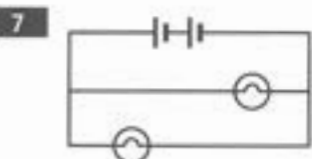
You press on the switch.



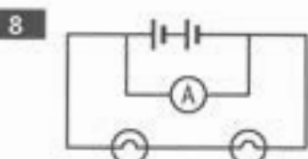
You add another battery.



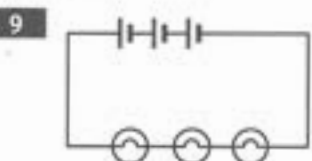
You add this voltmeter to the circuit.



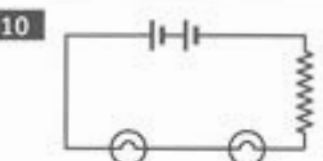
You unscrew one globe.



You add this ammeter to the circuit.



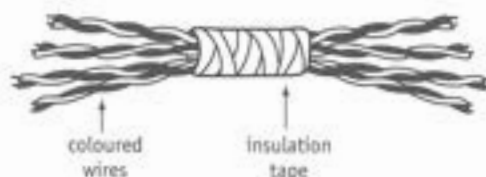
You add another globe.



You add a resistor to the circuit.



- 1** A bundle of several coloured wires, all insulated from each other, is taped together with black insulation tape. Unfortunately one of the wires is broken.



Design an experiment to find out which coloured wire is broken.

Title: _____

Aim: _____

Equipment: _____

Method: _____

Diagram:

Conclusion: _____

A student has built a quiz box, which is a question and answer game. This is a small box, which lists six questions on the left side and six answers on the right side (see box below). The end of a wire is connected to a contact behind each of the questions. The other end of the wire goes to a bell and a battery, both connected in series. Another wire comes off the battery and is connected to a contact in front of the corresponding answer. If a player selects the correct answer, the bell rings. If the selected answer is incorrect, nothing at all happens.

- 2** Match the questions on the left side of the box to the correct answers on the right side.

Questions	Answers
a Potential difference is measured in ... -	- ... amps
b Resistance is calculated by ... -	- ... amps
c Ohm's Law is about ... -	- ... dividing volts by amps
d The unit of current is ... -	- ... resistance
e A fuse melts when there are too many ... -	- ... volts
f House mains provide 240 ... -	- ... volts



- 3** Show what the game would look like in action by sketching the bell, battery, wires and box.

- 4** The student who built the quiz box had connected the contacts of the questions with contacts of the answers with wires inside the quiz box. Add those wires to your sketch of the box in question 2.
- 5** Design a similar question and answer box. Write your questions and answers in the table below. Then swap your design with someone in the class and figure out the wiring inside each other's quiz box by connecting the questions to the correct answers.

Questions		Answers



A science teacher gave the class (which was divided into six groups) the following instructions about an electricity experiment:

'Today we are going to do an experiment. Each of the six groups will do one part. At the end we are going to pool our results and figure out a conclusion. You will be performing the experiment twice, so these three groups — I'll call you Lot 1 — will be doing the same thing as those three groups — Lot 2. Okay. From both lots, each group is going to collect one piece of nichrome wire of a different length. Each group will make a different coil by winding the wire around a test tube: the first group's coil will have 5 turns, the second group's 10 turns, and the third group's 15 turns. Once you have the coil, put away the test tube and place the coil in a beaker with 100 mL of water. Connect the coil to a powerpack and turn the powerpack to its maximum setting. Your job is to time how long it takes for the water to boil. Yes, to boil, because you have just built a simple water heater. Off you go, and be careful.'

The student groups performed the experiment carefully. Here are their results:

Lot 1 measured 10 minutes for the short coil, 6 minutes for the middle-sized coil and 3 minutes for the long coil. Lot 2 measured 7 minutes for the short coil, 4 minutes for the middle-sized coil, and 2 minutes for the long coil. All the results were recorded onto the board. But the teacher was puzzled, very puzzled.

1 State a possible the aim for the experiment.

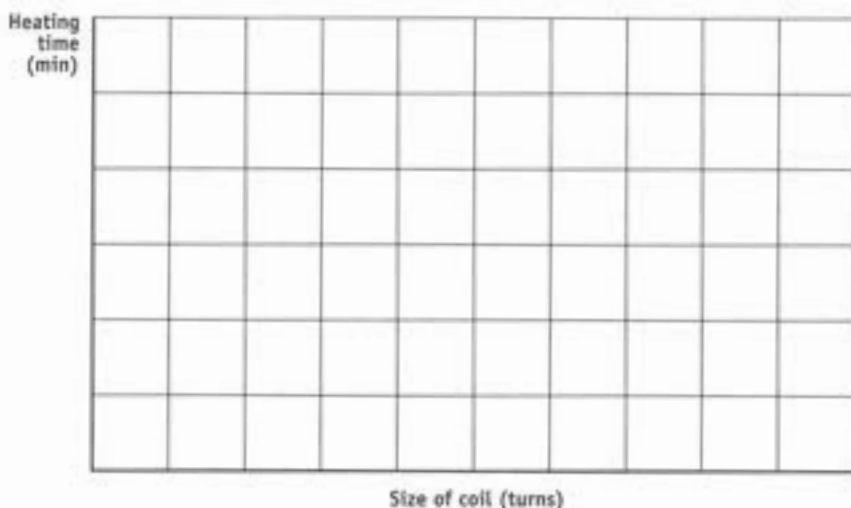
2 Write the method in the standard format.

3 Make a sketch of the experiment of one group.



- 4 Record the results for the two sets of three coils in a table.

- 5 Produce one graph showing both sets of results. The first set could be drawn by a blue line in your graph, the second set by a red line.



- 6 Suggest a conclusion for the experiment of one lot of students.

- 7 Why do you think the teacher wanted to do the same experiment twice?

- 8 Explain why the sets of results of the two lots were so different.

- 9 Suggest how the teacher could have avoided obtaining the two different sets of results.

When learning a new scientific concept, it can sometimes be difficult to understand what is going on — especially when you cannot actually see it. To help you understand, a simple model can be invented to explain what is happening. However, models do have their limitations and may need to be changed as your understanding becomes better.

In a circuit the power supply—the battery or the powerpack—acts as an energy bank, supplying the electrons with the same amount of energy as they travel. As the electrons move around the circuit, they use some of this energy.

Some materials are more difficult for electrons to move through than other materials. Such materials cause resistance in a circuit. Materials with low resistance allow the current to pass through easily; they are good conductors. Materials with high resistance do not allow much current to flow through; they are resistors. Sometimes the shape and length of the wire affects resistance, for example a thin piece of wire has a higher resistance than a thicker piece of wire.

Below is a model of an electric circuit, called the Human Circuit, which was performed by Ms Smart's class.



1 Look at the picture above and then complete the following table.

- Match the symbols with their names by joining the appropriate boxes with a line.
- Show which student or activity represents each symbol by joining the appropriate boxes with a line.
- Write the number of the activity in the last column.

Name	Sketch	Student or activity	Activity number
Ammeter		Climbing over a table	
Battery		Counting energy beans	
Globe		Giving out energy beans	
resistor		Pushing through a tube	



2 Answer these questions about the human circuit model.

- a What do the students going around the circle represent? _____

- b How would you include a switch in this model? _____

- c What represents the amount of voltage supplied? _____

3 In the space below design a **traffic circuit model**. Use any familiar traffic devices, for example cars, traffic lights, signs, roundabouts, etc. Remember to include a battery, switch, some kind of resistance, a globe and an ammeter. Compare your circuit with the rest of the class.

4 In your traffic circuit model what represents:

- a the electrons? _____
- b the amount of voltage supplied? _____

5 Compare your diagram to the rest of class. Write a list of common features.



In an electric circuit some electrical energy needs to be used up; that is, a resistance must slow the electric current, otherwise the unusually large amount of current travelling through the circuit will cause some of the circuit components to heat up, to melt or to spark. When components melt or get wet, electricity will flow along the path of least resistance, possibly causing a short circuit which often leads to fire. Fuses are designed to 'blow' before any item in a circuit causes a fire. When a fuse blows, the circuit is interrupted and electricity no longer flows.

Ammeters measure the current. Since they have a very low resistance themselves and do not influence the current, they are switched in series with the item to be measured. Voltmeters, on the other hand, measure the potential difference and do not want any current to flow through them. Therefore, a voltmeter has a very high resistance and is switched parallel with the item to be measured.

- 1** Summarise the information about ammeters and voltmeters by completing the table below.

Aspect	Voltmeters	Ammeters
Measures what?		
What does this measurement mean?		
Measures in what unit?		
Abbreviation for unit of measurement?		
Has what level of resistance?		
Is switched how?		
Is current allowed to travel through?		

Fixing electrical circuits

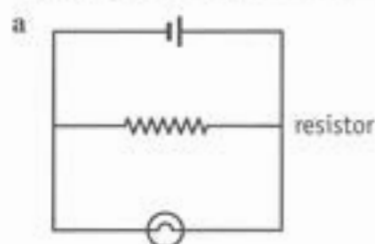
- 2** The following circuits all have something wrong with them, causing a short circuit. Describe what has caused the short circuit in each circuit and suggest, if possible, how to rectify the problem.

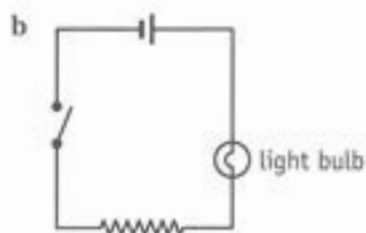
Circuit	Description of problem	Suggestion for rectification
a 		
b 		
c 		

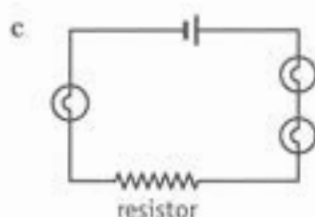
Circuit	Description of problem	Suggestion for rectification
d 		
e 		
f 		

Measuring electricity

3 Add ammeters and voltmeters to measure the current and potential difference of the indicated items. Describe each circuit in your own words.









Explaining Ohm's law

Ohm's law links current, potential difference and resistance. Current (I) is measured in amps (A), potential difference (U) in volts (V) and resistance (R) in ohms (Ω). According to Ohm's law we can calculate any one quantity, as long as we know the other two quantities. The law states that the resistance of an electrical DC device can be found by dividing the potential difference by the current.

- 1 Write Ohm's law as a formula, using the terms current, potential difference and resistance.

- 2 Write the law using the abbreviations I , U and R .

- 3 Express the unit of resistance, Ω , using the units of current and potential difference.

Manipulating Ohm's formula

Here is a hint to help you find the formula for calculating an unknown quantity: Cover the unknown quantity in the triangle with your finger. The position of the two remaining quantities in the triangle shows you whether you need to multiply them or divide them.



- 4 State the formula for finding:

- | | |
|-------------|------------------------------|
| a R _____ | b potential difference _____ |
| c U _____ | d resistance _____ |
| e I _____ | f current _____ |

Calculating using Ohm's law

- 5 Use Ohm's law to calculate the unknown in the following situations.

a What current passes through a lamp of resistance $60\ \Omega$ as a potential difference of $15\ \text{V}$ is applied across it? $R = 60\ \Omega$ $U = 15\ \text{V}$ $I = ?$ $I = U / R$ $I = 15 / 60 = 0.25$ The current measures $0.25\ \text{A}$.	b A $500\ \Omega$ resistor passes a current of $0.5\ \text{A}$. What potential difference is applied across it?
c A fuse will melt when a current of $8\ \text{A}$ flows through it. If the potential difference is $240\ \text{V}$, what is the lowest circuit resistance possible before the fuse will blow?	d A device in a small radio passes a current of $20\ \text{mA}$. The potential difference across that device is $6\ \text{V}$. What is its resistance?



Interpolating and extrapolating

One advantage of using a graph to record experimental results is that it can show all kinds of readings, even if they were not actually taken during the experiment.

2 When you *interpolate*, you are looking for answers between the highest and lowest recording. Look at your graph and find answers to the following questions.

- a How high would the amps be when the potential difference measures 4 V? _____
- b What amps match up to 7 volts? _____ 10 V? _____

3 When you *extrapolate*, you are looking for answers beyond the lowest or highest recording. Look at your graph and find answers to the following questions.

- a How high would the amps be, when the potential difference measures 20 V? _____
- b What current can be predicted to match 22 V? _____

Using Ohm's law

When Georg Ohm conducted a similar experiment in about 1830, he was puzzled that the answers produced such an excellent graph. He suspected that there was some kind of relationship that could be expressed by a mathematical formula. So Ohm began trying out different possibilities.

4 Now you follow the same steps that Ohm did to find the correct formula.

- a At first he added the volts and amps together. Perform the additions for the second and the third readings and compare the answers. _____

Does $V + A$ for the two settings provide the same answer? _____

- b In his second attempt Ohm multiplied the corresponding volts and amps. Perform the multiplications for the sixth and seventh settings. _____

Does $V \times A$ for the two settings give the same answer? _____

- c Ohm thought that dividing the amps into the volts might give him the desired answer. Divide the volts by the amps for the second and third settings. _____

Does $V \div A$ for the two settings produce the same answer? _____

- d Ohm would have been as excited as you are now, when he worked out which formula seemed to be right. Check for two more settings that your formula is indeed correct.

- e The answer, which is the same for any volt and ampere combination, is called the 'ohms' for that appliance (in honour of Georg Ohm). The ohms for any electrical appliance are calculated by using Ohm's law. All that is needed are the amps and the volts and the formula. Write down Ohm's law by writing the complete equation using ohms, volts and amps.

- f Ohms really are a measure of the resistance of an electrical appliance. Rewrite Ohm's law using current, resistance and potential difference.

Ohm's law allows us to calculate the resistance (= ohms) for any appliance as long as we have corresponding readings for the potential difference (= volts) and current (= amps).

5 Now let's apply Ohm's law to some electrical appliances in our homes. Many domestic appliances reveal a current rating on their label. You should already know what the potential difference is in your home. Calculate the resistance for the following appliances.

- a A radiator uses 10 amps. _____
- b A toaster draws 4 A. _____
- c An incandescent light bulb uses 0.25 A. _____
- d A fluorescent tube draws 0.1 A. _____



Ohm's law states that, for an electric conductor, the resistance measured in ohms is calculated by dividing the potential difference measured in volts by the current measured in amps.

Electric power measured in watts is calculated by multiplying the potential difference measured in volts and the electric current measured in amps.

If 1000 watts of electric power are used over 1 hour, 1 kilowatt-hour of energy has been used. The electricity supplier charges one-tenth of a dollar, or 10 cents, for each kilowatt-hour of electric energy used.

- 1** Join each word in the left column with its corresponding abbreviation in the right column by connecting them with a ruled line. One example has been done for you.

Word	Abbreviation
ohms	\$
volts	kWh
amps	h
watts	W
hour	A
kilowatt-hour	V
dollar	—
resistance	P
potential difference	t
current	E
power	I
energy	U
time	R
resistance formula $R =$	$E \times h \times \$0.10$
power formula $P =$	$P \times t$
energy formula $E =$	$I \times U$
cost formula $\text{cost} =$	U / I

Calculating with Ohm's law

- 2** Find the answers to the following problems. The first example has been done already.

- a An electric radiator is connected to mains electricity. It draws 5 A of current on its low setting. How high is its resistance?

Working out: Information: potential difference $U = 240 \text{ V}$ current $I = 5 \text{ A}$ resistance $R = ?$
 Formula: $R = U / I$
 Substitution: $R = 240 \text{ V} / 5 \text{ A}$
 Calculation: $240 / 5 = 48$
 Answer: The resistance of the heater measures 48 Ω .

Here is a handy hint: Cover the unknown quantity in the triangle to see how to find it.



Calculating electricity

15b

- b A television set uses 0.5 A. Calculate its resistance. _____

- c A small toaster has a resistance of 60 Ω . Calculate how much current it uses. _____

- d A tiny radio has a resistor in its circuitry that measures 1200 Ω and draws 0.005 A. How much potential difference does its battery provide? _____

- e Show your answers in the table below.

Appliance	Potential difference (V)	Current (A)	Resistance (Ω)

Calculating power and cost

First, develop your own equations for power P , energy E and cost C , using potential difference U , current I , time t and rate r .

 $P =$ $E =$ $C =$

Now use the units: W, kWh, \$, V, h.

 $W =$ $kWh =$ $\$ =$ $V =$ $h =$

- a A hair drier uses a current of 1.25 A. What is its power? _____
- b How much does it cost to run the hair drier for half an hour? _____

- c A vacuum cleaner uses a current of 3 A. What power does its motor provide? _____
- d It takes you 2 hours to clean the whole house with the vacuum cleaner from part c. How much electric energy are you using and how much does that energy cost?

- e A kettle uses a power of 1.2 kW for a time of 6 minutes. How much does that amount of energy cost?

- f Can you find the size of the current that the kettle in part e uses?

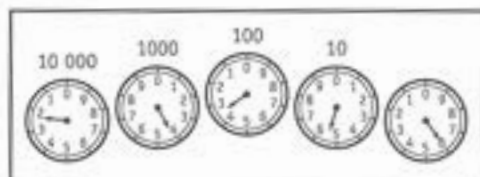
- g A clothes drier is rated as 2.4 kW. How much energy does it use in 1 h and what current does it draw?

- h How much does it cost to run the drier in part g for 2 h?

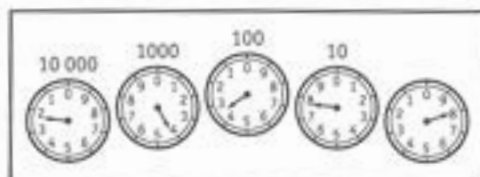
Problems at home

- 4 Kim's family is ready to go on holidays: the bags are packed, the lights are turned out and the children are strapped into their seat belts. Because her science class is covering the topic of

electricity, Kim decided to read the electricity meter before the family left. The dials on the meter showed 24 356 kWh. Ten days later Kim is surprised to discover that the reading of the electricity meter is different. Nobody has been living in the house while they were away on holidays, so why had the meter changed?



Electricity meter before holiday



Electricity meter after holiday

- What is the reading of the electricity meter after the holiday? _____
- How much energy had been wasted during the holiday period? _____
- How much money had been wasted on the empty house? _____
- How was the electricity used, when nobody had turned on any lights or appliances? _____
- If Kim had used a timer and left two 60 W lights on for 5 h every day, for security reasons, how much more would have been added to the wasted cost? _____
- Should Kim's family have turned off the main switch in the electricity box, so that no electricity was wasted? Why or why not? _____

5 All homes have fuses in their electricity box in order to protect people and property. A fuse for power points melts when a 15 A current is drawn at one time. Kim's parents are watching television, which uses 0.5 A, and they have a 1200 W heater switched on, as well as a 40 W lamp. Chris and Kim are studying in Kim's room, where they are using two 60 W desk lamps and another 1200 W radiator. Chris wants to study in another room but wonders whether it would be safe to switch on a 40 W lamp and a 900 W fan heater.

- Calculate whether the fuse would melt if Chris turned on the extra appliances in another room. _____
- It is possible in some houses to have several 2400 W heaters on without the fuse blowing. Suggest a way that would make this possible. _____
- Houses have 8 A fuses for lights, 15 A fuses for power points and 30 A fuses for electric stoves. Suggest reasons why there are different circuits for lights, power points and stoves and why their corresponding fuses have different amp ratings. _____
- At home locate your electricity box and count the number of fuses. If possible describe the different types of fuse wire, stating some similarities and some differences. _____



Yesterday at school, our science class did an experiment about electricity. The teacher gave us two small appliances (a light bulb and a buzzer) and told us that one of the appliances was an ohm resistor, but the other one was not. An ohm resistor obeys Ohm's law exactly. To find the answer, we decided to set up an electrical circuit with a DC powerpack, an ammeter, a voltmeter and one of the appliances. We then measured the potential difference and the current for eight settings on the powerpack. Here are the eight measurement pairs: 10 V and 2 A, 2 V and 0.4 A, 6 V and 1.2 A, 16 V and 3.2 A, 3 V and 0.6 A, 8 V and 1.6 A, 4 V and 0.8 A, 18 V and 3.6 A. We then replaced that appliance with the other one and repeated measuring amps and volts several times. Here are those results: 0 V and 0 A, 12 V and 5 A, 8 V and 2.5 A, 5 V and 1.5 A, 2 V and 0.2 A, 5 V and 1 A, 14 V and 6 A. The teacher said that if we made a graph of the results, the answer would become obvious. So we made a line graph and drew the two resulting lines onto it. Sure enough, the answer was right there in front of us.

Looking at the text

1 The text above is a recount; it retells the experiment the students had performed earlier. Suggest possible answers for the following questions.

- Who could be the intended audience of the text? _____
- Who could be the author? _____
- Why was the text produced? _____
- Was the text originally written or spoken? Give reasons for your answer.

Looking at the results

2 According to Ohm's law the resistance of an appliance stays the same, no matter what potential difference and current are applied to it. Resistance is calculated by dividing the volts by the amps, and the answer is given in ohms.

- Select any three pairs of measurements for each appliance used in the experiment and calculate the resistances.

Appliance 1: _____

Appliance 2: _____

- State your opinion as to which appliance is an ohm resistor. _____
- Justify your opinion. _____

Recording the experiment scientifically

3 a Rewrite the recount text above as an experimental record and include the students' graph and conclusion. Use the scaffold below.

Title: _____

Aim: _____

Equipment: _____

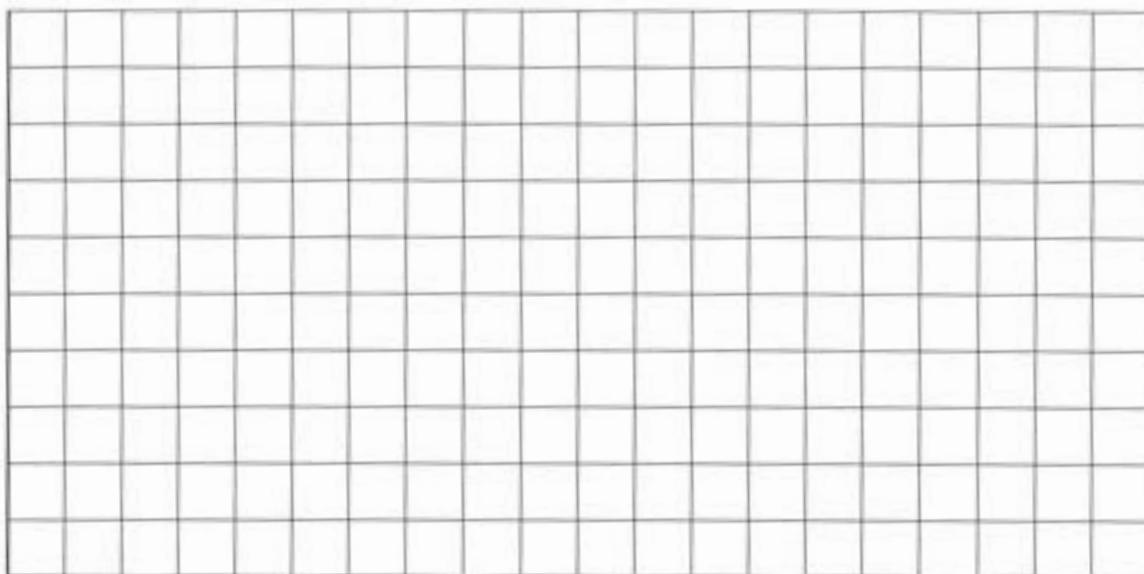


Method: _____

Diagram:

Results: _____

Current
(A)



Potential difference (V)

Conclusion: _____

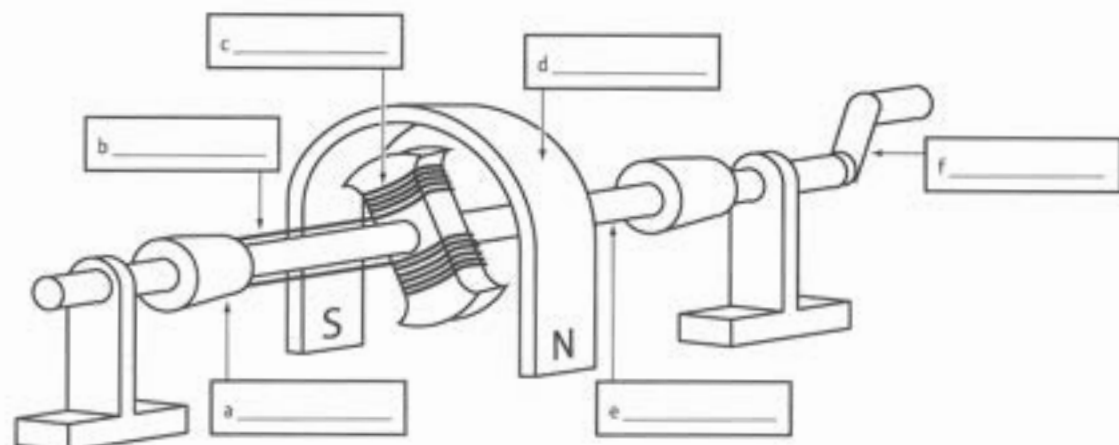
- b Look at the line for the ohm resistor in your graph and find the current for 7 V and 11 V.

- c Extrapolate your ohm resistor graph and suggest values for 0 V and 20 V.

If we could change electrical energy into movement energy, then we could make things move or turn as long as we had electricity. And if we could turn movement energy into electrical energy, we could use electricity as long as we moved or turned something. An electric motor and an electric generator do exactly those jobs, and life has been much easier for people because of them.

The electric generator

1 The parts of an electric generator are listed in the 'Label' column of the table below. Refer to that list to label the following diagram of an electric generator.



2 Join the boxes to match the labels with the correct descriptions and their matching functions in the table below.

Label	Description	Function
Axle	Connect coil wire with two wires to a light bulb.	Allows current to flow through it.
Coil	Contains many loops of wire.	Holds the coil and handle and is able to spin easily.
Coil wire	Is a long round rod.	Makes an electric connection between the coil wire and a light bulb and so lets the bulb shine.
Contacts	Is a long, thin and insulated wire.	Makes the coil spin inside the horseshoe magnet.
Handle	Is a magnet in the shape of a horseshoe.	Places a south pole and a north pole on either side of the coil, thus creating a magnetic field.
Horseshoe magnet	Is connected to one end of the axle.	When it moves through a magnetic field, electrons move through its wire, ie an electric current flows.

3 Answer these quick quiz questions about the electric generator:

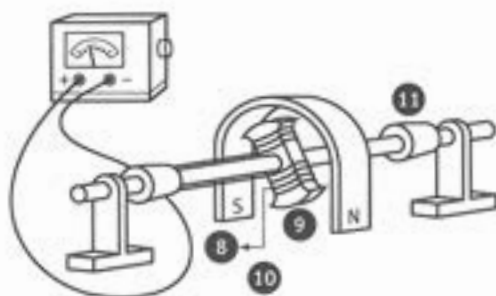
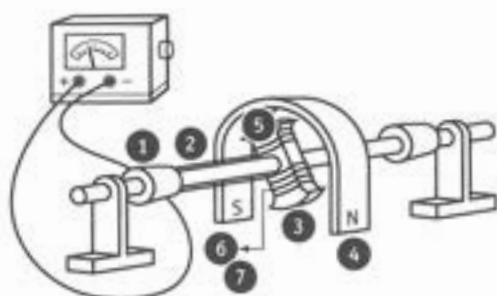
- How many electric wires go to a possible light bulb? _____
- How many poles are there in a magnet? _____
- How many coil wires are there? _____
- How many contacts are there? _____
- What are at the ends of the horseshoe magnet called? _____
- What is produced when electrons move? _____

- g What three things are connected to the axle? _____
- h What moves when the handle is turned? _____
- i What does a light bulb do when current flows? _____
- j What is an electric current? _____

The electric motor

- 4 Refer to the terms in the box below and label the diagram of an electric motor on the right.

power source horseshoe magnet electric wire contacts coil wire coil axle



- 5 Write the numbers from the diagrams above next to the corresponding statements below.

- | | | |
|-------|---|--|
| _____ | a | Any device connected to the spinning axle would also move. |
| 8 | b | As soon as the coil has moved half a turn, the current flows through the coil in the opposite direction. |
| 2 | c | Electricity flows through the coil in one direction. |
| _____ | d | Electric power is supplied through the wires to the contacts. |
| _____ | e | The coil becomes an electromagnet with a north pole at one end and a south pole at the other. |
| _____ | f | The coil continues to spin. |
| _____ | g | The coil ends move away from the magnet. |
| _____ | h | The coil turns around the axle. |
| _____ | i | The electromagnetic north pole repels the north pole of the horseshoe magnet. |
| _____ | j | The electromagnetic south pole also repels the south pole of the horseshoe magnet. |
| _____ | k | Again the poles of the electromagnetic coil repel the horseshoe magnet. |



These days we hear quite a lot about building energy-efficient houses and insulating houses in order to reduce the amount of electricity we use. We are also being educated about how we can use less electricity at home. There is also continuing research into alternative sources of electricity, although most of our electricity is produced from coal. Should we continue producing electricity from coal? In this activity you are going to discuss the advantages and disadvantages of coal-electricity and then draw your own conclusion.

The following information about electricity production from coal has been collected and placed into three boxes.

A. Explanation of coal-electricity:

- Black coal is burnt.
- Black coal is mined.
- Heat boils water into steam.
- Steam turns a turbine.
- Turbine changes movement into electricity.

B. Arguments for coal-electricity:

- Australia exports about half of its coal to Japan and Europe.
- Coal can be mined in all weather conditions.
- Coal deposits are near the surface of the earth.
- Easy to extend network to new areas.
- Electricity from coal generators is reliable.
- Electricity is not dependent on weather or cycles.
- Enough coal for 400 years of electricity production.
- Established network of coal generators and electric cables.
- Large coal deposits in Australia.
- Network is working well at present.
- No tunnels and lifts needed as in underground mines.
- Open-cut mining is cheap.
- Open-cut mining is easy.

C. Arguments against coal-electricity:

- Acid rain damages plants and weathers buildings.
- Alternative ways to produce electricity need to be researched, which is expensive.
- It is expensive to test and improve alternative electricity production methods.
- Changes in temperature and chemicals disturb the natural balance of the environment.
- Coal burning is responsible for nearly 50% of Australia's carbon dioxide emission.
- Coal is washed with water from lakes and rivers.
- Dirty water goes back but is now polluted with chemicals.
- Global warming will change weather patterns and raise the ocean levels.
- Habitats are reduced and partly destroyed.
- Increased carbon dioxide in the atmosphere leads to global warming.
- Large areas of land are necessary for mining of coal.
- Many natural bush areas and fields are used up for electric cables and electric towers.
- Much land is used for coal power stations.
- Nitrogen oxides and sulfur dioxide from burning coal dissolve in raindrops.
- Non-renewable resource will run out eventually.
- Power plant equipment is cooled with river and lake water.
- Raindrops contain nitric acid and sulfuric acid.
- 1 MWh (= megawatt hour) of electricity produces nearly 1 ton of carbon dioxide.

- 1 Arrange the points into order by writing numbers in front of them: box A is about how electricity is produced from coal; box B provides four main arguments in favour of coal-electricity; box C contains five main arguments against coal-electricity.

Electricity from coal?

18b



- 2** You are now going to write your discussion. First introduce the issue of coal-electricity and explain it. Then make it clear that there are arguments in favour of coal-electricity and arguments against it. Write the introductory paragraph, referring to your first group of ordered pieces of information. Use the scaffold below and the hints at the end to help you.

Introduction:	
Statement of issue	
Explanation of issue	
Preview of arguments	

Now write the section supporting coal-electricity. There are four arguments in its favour. For each argument there should be a main point (ie *what is this argument about?*) and an elaboration (ie *what explains or supports this argument?*). Use your ordered statements from box B above and the scaffold below to write four arguments in favour of coal-electricity.

Arguments for:	
Point 1: Size of coal deposits Elaboration 1	
Point 2: Open-cut mining Elaboration 2	
Point 3: Reliable electricity Elaboration 3	
Point 4: Network of coal-electricity Elaboration 4	

Now write the section which is against coal-electricity. There are five arguments against it. Again use the structure of point and elaboration to construct each argument. Use your ordered statements from box C above and the scaffold below to write five arguments against coal-electricity.



Arguments against:	
Point 1: Non-renewable resource	
Elaboration 1	
Point 2: Greenhouse gas emission	
Elaboration 2	
Point 3: Acid rain gas emission	
Elaboration 3	
Point 4: Destruction of land areas	
Elaboration 4	
Point 5: Thermal and chemical pollution	
Elaboration 5	

Finally, you must write a conclusion. Decide whether you are in favour of coal-electricity or against it, and write down your opinion. Note that there is no right or wrong answer, but it is important that you are able to justify your point of view. Justify your opinion by summarising the most important arguments. To finish, you may like to recommend what you think ought to happen in the future. In your conclusion you need to sound convincing: use emotive words (eg *terribly important*, *widespread devastation*, *must*, *should never*) and exaggerate a little (use words like *always*, *rarely*, *obviously*, *everybody*). Write your conclusion in the scaffold below.

Conclusion:	
Opinion	
Summary of key arguments	
Recommendation	

Handy hints:

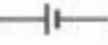
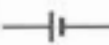
- Statement of issue = what is this discussion about?
- Explanation of issue = how is electricity produced from coal?
- Preview of arguments = what are the arguments about?
- Point = what topic is this paragraph about?
- Elaboration = what information is there to support or explain the topic?
- Opinion = which side do you think is right?
- Summary of key arguments = what are the most important arguments supporting your view?
- Recommendation = what should happen now?



Below is a domino game with the symbols, names, sketches, diagrams and formulas from the topic of electricity.

Instructions:

- Cut out the electrino pieces (each piece is two squares).
- Place them face down on the table.
- Divide the pieces between each of two to four players. Players look at their pieces.
- One piece is placed face up in the middle of the table. Each player in turn joins one piece onto either end of the pieces on the table, making a long chain. You can match up corresponding symbol, name, formula, diagram or sketch. For example, to 'U' you could add any of these: potential difference, R I, volts, V, voltmeter, sketch of voltmeter, diagram of voltmeter.
- The first person to use all their pieces is the winner.

Ω	potential difference	volts	battery	amps		hours		watts	
kilowatt hours	$= U \cdot I$	dollars	P	current	U		kWh	energy	t
time		W	P . h		resistance	V	A	I	
h	cost	U / I			R	resistor	P . t . \$0.15	R I	
ammeter	E	\$		lamp	voltmeter		wire		power
resistor	U / R		volts	V	fuse		switch	A	
fuse	amps	Ω	W	ohms		ohms			U



1 Electric history

- (1) Rubbed amber attracts sawdust and magnetite always points in the same direction.
(2) Conductors allow positive or negative electricity to flow and insulators do not.
(3) Ohm's law is about electric pressure, electric current and electric resistance.



3

Name	Date	Country	Contribution
Thales of Miletus	530 BC	Greece	Rubbed amber attracts sawdust.
Chinese investigators	300 BC	China	Magnetite spoons always point in the same direction (magnetic compass).
Girolamo Cardano	1750	Italy	Electrostatic attraction and magnetic attraction are different.
Stephen Gray	1729	England	Conductors let electricity flow, insulators do not.
Charles Dufay	1733	France	There are two kinds of electricity.
Benjamin Franklin	1740	America	Electricity is one kind of fluid which flows from positive to negative.
Georg Ohm	1827	Germany	Ohm's law is about measuring electricity in volts, amps and ohms.

- 4 Longest: Benjamin Franklin, 84 years. Shortest: Charles Dufay, 41 years.
- 5 a was, experimented, is, occurring, are, is made, is noticed, were attracted, stuck, had been rubbed, were experimenting, called, were able to spin, pointed, became, seemed to be, was, attracted, attracted, performed, showed, were, was distinguishing
- 6 discovered, could move, tried, found, flow, did, called, stated, was, polished, learned or learnt, repelled, did, attracted, concluded, were, conducting, believed, was, defined, stated, flowed

7

Action verb	Linking verb	Saying verb	Thinking verb
build	is made up of	shout	feel
experimented	was, is, were	called	noticed
is made	is called	stated	found
were attracted	seemed to be		learned

- 8 In 1927 the results intrigued German scientist Georg Ohm (1789–1854) when electricity was measured by him. We now know this relationship as 'Ohm's law'.

2 Whose bright ideas?

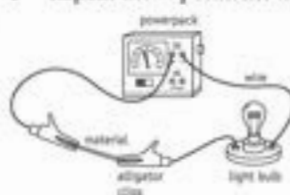
Name	Date	Country	Contribution
Michael Faraday	1828	England	Built electric motors and dynamos.
Joseph Henry	1828	America	Built a very strong electromagnet.
Georg Ohm	1827	Germany	Discovered Ohm's Law: resistance = potential difference / current.
William Sturgeon	1825	England	Built the first electromagnet.
Andre Ampere	1820	France	Noticed that a coil was magnetic.
Hans Christian Oersted	1820	Denmark	Linked electricity and magnetism.
Alessandro Volta	1796	Italy	Built the first electric battery.
Luigi Galvani	1786	Italy	Frog leg muscles work with electricity.
Benjamin Franklin	1752	America	Discovered thunderstorms were electrical.
Thales of Miletus	530 BC	Greece	Showed that amber attracted sawdust.

3 Conductors and insulators

- In order: Aim, Equipment, Method, Diagram, Results, Conclusion
- The aim was to see which materials conduct electricity.
- Apparatus: powerpack, light bulb, wires (3), alligator clips (2)
Materials: small pieces of copper, wood, aluminium foil, plastic, carbon, cardboard, iron, rubber, cotton, string
- a Connect one wire between the positive terminal of a powerpack and a light bulb, another wire between the light bulb and an alligator clip and a third wire between the negative terminal of the powerpack and an alligator clip.
b Check the electric circuit.

- Place a piece of material between the alligator clips.
- Turn the electricity on and observe.
- Turn the electricity off.
- Repeat the experiment with all the pieces of material.

5



- The light bulb shone brightly with copper, aluminium, carbon and iron. But it did not come on with plastic, cardboard, rubber, cotton and string.
- Metals and carbon conduct electricity and non-metals do not.

8

Materials	Observations	Conclusions
copper, aluminium	light came on	conductor
carbon, iron	light came on	conductor
plastic, cardboard	light stayed off	insulator
rubber, cotton	light stayed off	insulator
string	light stayed off	insulator

- A conductor allows electricity to flow through it.
- An insulator stops electricity from flowing through it.
- Plastic is an insulator, therefore you cannot receive an electric shock when you touch the plastic surrounding the wire while electricity is flowing through the wire. The electricity can only flow along the wire.
- Gold is a much better conductor than copper, therefore computers work more efficiently and are also smaller if they contain gold wires.

4 Circuit symbols

- They are both associated with electricity; they both form a double loop.
 - B is much simpler and easier to read. It uses symbols not sketches.
- device, supplier, user, controller, controller, device
- It is faster and simpler to draw, easier to read, less cluttered.
- Teacher to check.

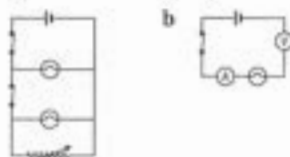
2

Symbol	Group
	measuring devices
	energy suppliers
	energy users
	circuit controllers
	circuit controllers
	measuring devices

5 Series and parallel circuits

- in series
 - parallel
 - parallel
 - in series
- B
 - C
 - A

3



- A battery powers 2 light bulbs in series and a third one switched in parallel to one light bulb. The third bulb has a switch after it.
 - A battery supplies electricity to 4 light bulbs in parallel with 4 switches.
- In a series circuit the flow of electrons to all bulbs is interrupted when one light bulb goes out and so none of the lights work any more. In a parallel circuit the electrons can use an alternative route, which is not interrupted by the broken light bulb, and so the other lights continue to work.
- in series
 - four
 - Teacher to check
 - nothing
 - Globe 2 would be even brighter.
 - nothing
- parallel
 - 4 (when switches are closed)
 - Each globe is the same brightness.
 - Globe 2 because of open switch.
- Houses would be wired as in question 7, because this way lights are not affected when other lights are turned on or off.

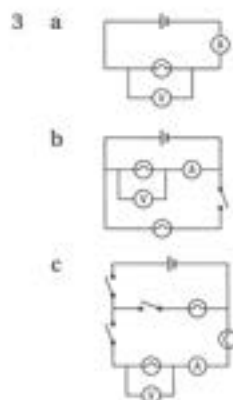
6 Ammeters and voltmeters

- Current is measured in amperes, whereas potential difference is measured in volts.
 - Ammeters have low resistance and should not be connected straight across the power supply, however voltmeters have high resistance and can be connected straight across the power supply.
 - Large amounts of current flow through an ammeter, yet only small amounts travel through a voltmeter.
 - Current stays the same in a series circuit, while potential difference varies for different appliances.
 - In an ammeter, energy is used up by the lamp, whereas energy is used up around the circuit in a voltmeter.
 - Ammeters are switched in series next to the appliance, but voltmeters are placed in parallel across the appliance.
- The ammeter should be placed before or after the light bulbs.
 - The voltmeter should be parallel to the resistor.
- i) voltmeter ii) ammeter iii) ammeter
 - i) 4 V ii) 2 A iii) 2 A
- 0.7 A
 - 1.8 A
 - 1.5 A

7 Circuits in text, sketches and diagrams

1 Note: Letters refer to the original positions in the table

Name	Sketch	Symbol	Function
a AC power source	g	j	h
b ammeter	h	g	e
c DC power source	c	b	i
d fuse	i	i	g
e light bulb	a	f	b
f resistor	b	h	j
g rheostat	j	c	d
h switch	f	a	c
i voltmeter	d	e	f
j wire	e	d	a



2

Text	Sketch	Diagram
a An electric circuit is made up of three items connected by wires. The positive terminal of a powerpack is connected with a wire to a light bulb. The bulb is connected to a switch and the switch to the negative terminal.		
b A 6V battery is connected to two light bulbs in parallel.		
c A battery, two globes and a switch are connected in series.		
d This circuit has six items connected to a powerpack by wires. Two light bulbs are connected in series and then two more light bulbs are connected in parallel. There is one switch before each parallel bulb.		
e A powerpack is connected to a light bulb, a rheostat and an ammeter, all in series.		
f A battery, switch, light globe and ammeter are switched in series. Parallel to the battery is a voltmeter.		

8 Changing circuits

- 1 Batteries will counteract each other quickly, resulting in no lights on.
- 2 All 3 globes will be the same brightness.
- 3 Both globes will go out.
- 4 It's a short circuit – globes will go out and batteries empty quickly.
- 5 Volts are increased, therefore brightness will increase.



- 6 Voltmeter is incorrectly switched in series and will stop the current, light will go out.
- 7 Unscrewed globe will go out, but will have no effect on other globe.
- 8 Short circuit through ammeter, which will burn out. Lights will go out.
- 9 Globes will become dimmer.
- 10 Globes will become dimmer.

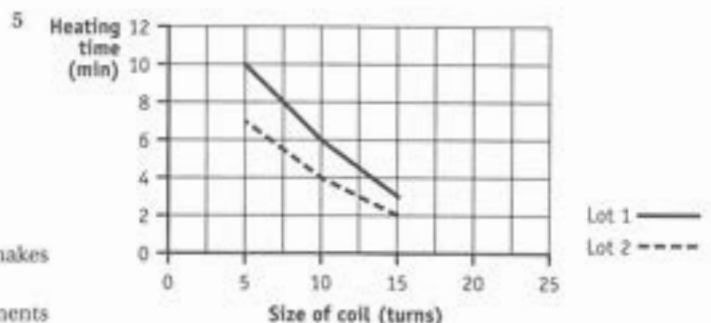
9 Solving problems electrically

- 1 Connect a light bulb and a powerpack to the ends of each coloured wire, one at a time. All the light bulbs will light up except the one attached to the broken wire.
- 2 a volts
d amps
b dividing volts by amps
e amps
c resistance
f volts
- 3–5 Teacher to check.

10 Heating water

- 1 To find whether the length of a coil is important in heating water.
- 2 ■ Wind a length of nichrome wire around a test tube so there are five turns in the coil of wire.
■ Remove the test tube.
■ Submerge the coil in 100 mL of cold water.
■ Connect the coil to a powerpack on maximum setting.
■ Turn the power on and time how long it takes for the water to boil.
■ Repeat the experiment with a 10-turn coil and a 15-turn coil.
- 3 Teacher to check.

Number of turns in coil	Time of lot 1 (min)	Time of lot 2 (min)
5	10	7
10	6	4
15	3	2



- 6 A coil with more turns heats up water faster.
- 7 Repeating an experiment and getting similar results makes the results more conclusive.
- 8 The two student lots must have performed the experiments slightly differently, so the results could not be compared accurately. Lot 1 may have used a larger test tube than Lot 2, or may have had colder water to begin with, for example.
- 9 The teacher needs to be more specific in the instructions, i.e. use a small test tube to make the coils and record the temperature of the water before heating.

11 Circuit models

Name	Symbol	Activity	Activity number
Globe		Climbing over a table	3 and 6
Ammeter		Counting energy beams	5
Resistor		Pushing through a tube	4
Battery		Giving out energy beams	1

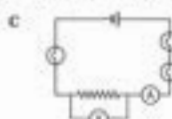
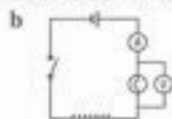
- 2 a electrons
b Include a door that can be open or closed.
c The difference in number of lollies before and after the activity run.
- 3 Teacher to check.

12 Short circuits and fuses – Part 1

Aspect	Voltmeters	Ammeters
Measures what?	potential difference	current
What does this measurement mean?	electrical pressure between two points	number of electrons
Measures in what unit?	volts	amperes
Abbreviation for unit of measurement?	V	A
Has what level of resistance?	high	low
Is switched how?	parallel	in series
Is current allowed to travel through?	no	yes

- 2 a The powerpack is short circuited and its fuse will blow. Should include light bulb or resistor.
b Since the ammeter has a very low resistance, the powerpack is short circuited and will damage the ammeter, probably blowing its fuse. Add a light bulb or resistor.

- c The voltmeter has a very high resistance and a current will not flow. Do not use the voltmeter or switch it in parallel.
 d The current will be shared between the ammeter and the bulb; the ammeter is not measuring the current through the bulb and will be damaged as it has formed a short circuit. Switch the ammeter in series with the bulb.
 e The voltmeter will not allow any current to travel through the bulb because of its high resistance. Switch the voltmeter in parallel with the bulb. The ammeter will blow. Switch it in series not in parallel.
 f The voltmeter will not allow any current to flow. Switch the voltmeter in parallel with the bulb. The ammeter is correct.



13 Short circuits and fuses – Part 2

- 1 resistance = potential difference / current

2 $R = U / I$

3 $\Omega = V / A$ (ie volts per ampere)

4 a $R = U / I$

b potential difference = current $\times$ resistance

c $U = I \times R$

d resistance = potential difference / current

e $I = U / R$

f current = potential difference / resistance

5 a $I = 15 / 60 = 0.25 \text{ A}$

b $U = 500 \times 0.5 = 250 \text{ V}$

c $R = 240 / 8 = 30 \Omega$

d $R = 6 / 0.020 = 300 \Omega$

e $I = 12 / 48 = 0.25 \text{ A}$, no

f $I = 12 / 24 = 0.5 \text{ A}$, no

g $R = 36 \Omega$, $U = 12 \text{ V}$, $I = 12 / 36 = 0.33 \text{ A}$

h $R = 12 / 0.5 = 24 \Omega$

- 6 Replace the broken fuse wire.

- 7 Reset the circuit breaker by pressing the spring back.

14 Ohm's law rediscovered

- 1 Aim: To measure corresponding measurements for current and potential difference.

Equipment: powerpack, rheostat, ammeter, voltmeter, wires (5)

Method:

- Make an electrical circuit and switch a powerpack, a rheostat and an ammeter in series, and a voltmeter parallel to the rheostat.
- Move the slide of the rheostat into any position and turn on the power.
- Record the potential difference and the current.
- Repeat the readings for six more settings on the rheostat.

Conclusion: As the potential difference increases over a rheostat, the current also increases steadily but slower.

- 2 a 2 A b about 3.5 A; 5 A

- 3 a 10 A b about 11 A

- 4 a $2 + 1 = 3$; $5 + 2.5 = 7.5$; no

- b $15 \times 7.5 = 112.5$; $18 \times 9 = 162$; no

- d $8 / 4 = 2$; $12 / 6 = 2$

- e ohms = volts / amps

- c $2 / 1 = 2$; $5 / 2.5 = 2$; yes

- f resistance = potential difference / current

- 5 a resistance of radiator = 240 volts / 10 amps = 24 ohms

- b resistance of toaster = 240 volts / 4 amps = 60 ohms

- c resistance of light bulb = 240 volts / 0.25 amps = 960 ohms

- d resistance of tube light = 240 volts / 0.1 amps = 2 400 ohms

Results:

Setting	Pot. diff. (V)	Current (A)
1	0	0
2	2	1
3	5	2.5
4	8	4
5	12	6
6	15	7.5
7	18	9

The line in the graph is a straight line starting at the origin and going towards the top right corner.

15 Calculating electricity

- 1 ohms – Ω , volts – V, amps – A, watts – W, hour – h, kilowatt-hour – kWh, dollar – \$, resistance – R, potential difference – U, current – I, power – P, energy – E, time – t, $R = U / I$, $P = I \times U$, $E = P \times t$, cost = $E \times h \times \$0.10$

- 2 b $R = U / I = 240 \text{ V} / 0.5 \text{ A} = 480 \Omega$

- c $R = U / I$ therefore $I = U / R = 240 \text{ V} / 60 = 4 \text{ A}$

- d $R = U / I$ therefore $U = R \times I = 1200 \Omega \times 0.005 \text{ A} = 6 \text{ V}$

e

Appliance	Potential difference (V)	Current (A)	Resistance (Ω)
television	240	0.5	480
toaster	240	4	60
radio	6	0.005	1200

- 3 a Power of hair drier = $P = U \times I = 240 \text{ V} \times 1.25 \text{ A} = 300 \text{ W} = 0.3 \text{ kW}$

- b Cost of hair drier = $P \times t \times \$0.10 = 0.3 \text{ kW} \times 0.5 \text{ h} \times \$0.10/\text{kWh} = \$0.015 = 1\frac{1}{2}$ cents

- c Power of cleaner = $P = U \times I = 240 \text{ V} \times 3 \text{ A} = 720 \text{ W} = 0.72 \text{ kW}$



- d Cost of cleaner = cost = $P \times t \times 0.10 = 0.72 \text{ kW} \times 2 \text{ h} \times \$0.10/\text{kWh} = \$0.14 = 14 \text{ cents}$
 Energy = $E = P \times t = 0.72 \text{ kW} \times 2 \text{ h} = 1.44 \text{ kWh}$
 e Cost of kettle = cost = $P \times t \times 0.10 = 1.2 \text{ kW} \times 6/60 \text{ h} \times \$0.10/\text{kWh} = 1.2 \times 0.1 \times 0.10 = \$0.012 = 1 \text{ cent}$
 f Power of kettle = $P = U \times I$ therefore $I = P / U = 1200 \text{ W} / 240 \text{ V} = 5 \text{ A}$
 g Energy of drier = $E = P \times t = 2.4 \text{ kW} \times 1 \text{ h} = 2.4 \text{ kWh}$; $P = U \times I$, therefore current = $I = P / U = 2400 \text{ W} / 240 \text{ V} = 10 \text{ A}$
 h Cost of drier = $P \times t \times 0.10 = 2.4 \text{ kW} \times 2 \text{ h} \times \$0.10/\text{kWh} = \$0.48 = 48 \text{ cents}$
- 4 a 24 378 kWh b 24 378 kWh - 24 356 kWh = 22 kWh c 22 kWh $\times$ $\$0.10 = \2.20
 d Eg: the fridge was still on, so were the TV and stereo, both controlled by remote, the clock radio etc.
 e $60 \text{ W} = 0.060 \text{ kW}$ energy = $0.060 \text{ kW} \times 2 \text{ globes} \times 5 \text{ h} \times 10 \text{ days} = 6 \text{ kWh}$ cost = $6 \text{ kWh} \times \$0.10 = \0.60
 f Eg: Yes, so there is no possibility of electric short circuit and fire; No, because house is obviously empty so there is greater risk of burglary.
- 5 a Television: 0.5 A ; heater: $P = I \times U$, therefore $I = P / U = 1200 \text{ W} / 240 \text{ V} = 5 \text{ A}$;
 Lamp: $I = P / U = 40 \text{ W} / 240 \text{ V} = 0.17 \text{ A}$;
 Kim's room: total power = $60 \text{ W} \times 2 + 1200 \text{ W} = 1320 \text{ W}$ $I = P / U = 1320 \text{ W} / 240 \text{ V} = 5.5 \text{ A}$;
 Total current so far: $0.5 \text{ A} + 5 \text{ A} + 0.17 \text{ A} + 5.5 \text{ A} = 11.17 \text{ A}$;
 Chris' room: $900 \text{ W} + 40 \text{ W} = 940 \text{ W}$ $I = P / U = 940 \text{ W} / 240 \text{ V} = 3.91 \text{ A}$;
 Total current: $11.17 \text{ A} + 3.91 \text{ A} = 15.08 \text{ A}$, which is more than the fuse allows, so the fuse will melt.
 b Having several circuits (each with a 15 A limit) would allow several heaters as long as they are on different circuits.
 c The power used by lights is fairly low and so a lower fuse (which is safer) can be used. Power points draw more current because of heaters, air conditioners etc and are limited to 15A. Stove use a lot of power (four hotplates would be like four heaters!) and so require a higher fuse rating.
 d Answers will vary.

16 Resistors

- 1 a friends, parents, other science students
 b science students at school
 c to recount an event from school
 d Written in a recount (for example to read in class) or a letter, as there is a lot of detail and it is reasonably formal.
- 2 a Eg: $R = 10 \text{ V} / 2 \text{ A} = 5 \Omega$, $R = 2 \text{ V} / 0.4 \text{ A} = 5 \Omega$ or $R = 12 \text{ V} / 5 \text{ A} = 2.4 \Omega$, $R = 8 \text{ V} / 2.5 \text{ A} = 3.1 \Omega$
 b appliance 1
 c The resistance stays the same for the various settings.
- 3 a approx. 1.4 A, approx. 2.2A
 b 0 A, 4 A

17 Movement and electricity

- 1 a contacts b coil wire c coil d horseshoe magnet e axle f handle

Label	Description	Function
Axle	Is a long round rod.	Holds the coil and handle and is able to spin easily.
Coil	Contains many loops of wire.	When it moves through a magnetic field, electrons move through its wire, ie an electric current flows.
Coil wire	Is a long, thin and insulated wire.	Allows current to flow through it.
Contacts	Connect coil wire with two wires to a light bulb.	Makes an electric connection between the coil wire and a light bulb and so lets the bulb shine.
Handle	Is connected to one end of the axle.	Makes the coil spin inside the horseshoe magnet.
Horseshoe magnet	Is a magnet in the shape of a horseshoe.	Places a south pole and a north pole on either side of the coil, thus creating a magnetic field.

- 3 a 2 b 2 c 1 d 2
 e north pole and south pole f electricity or current g handle, coil, contacts
 h axle and coil i lights up j moving electrons
- 5 a 11 b 8 c 2 d 1 e 3 f 10
 g 6 h 7 i 4 j 5 k 9

18 Electricity from coal?

- 1 A. Explanation of coal-electricity:

- Black coal is mined.
- Black coal is burnt.
- Heat boils water into steam.
- Steam turns a turbine.
- Turbine changes movement into electricity.



B. Arguments for coal-electricity:

Size of coal deposits:

- Large coal deposits in Australia.
- Enough coal for 400 years of electricity production.
- Australia exports about half of its coal to Japan and Europe.

Open-cut mining:

- Coal deposits are near the surface of the earth.
- Open-cut mining is easy.
- No tunnels and lifts needed as in underground mines.
- Open-cut mining is cheap.

Reliable electricity:

- Electricity from coal generators is reliable.
- Electricity is not dependent on weather or cycles.
- Coal can be mined in all weather conditions.

Network of coal-electricity:

- Established network of coal generators and electric cables.
- Easy to extend network to new areas.
- Network is working well at present.

C. Arguments against coal-electricity:

Non-renewable resource:

- Non-renewable resource will run out eventually.
- Alternative ways to produce electricity need to be researched, which is expensive.
- It is expensive to test and improve alternative electricity production methods.

Greenhouse gas emission:

- Coal burning is responsible for nearly 50% of Australia's carbon dioxide emission.
- 1 MWh (= megawatt hour) of electricity produces nearly 1 ton of carbon dioxide.
- Increased carbon dioxide in the atmosphere leads to global warming.
- Global warming will change weather patterns and raise the ocean levels.

Acid rain gas emission:

- Nitrogen oxides and sulfur dioxide from burning coal dissolve in raindrops.
- Raindrops contain nitric acid and sulfuric acid.
- Acid rain damages plants and weathers buildings.

Destruction of land areas:

- Large areas of land are necessary for mining of coal.
- Much land is used for coal power stations.
- Many natural bush areas and fields are used up for electric cables and electric towers.
- Habitats are reduced and partly destroyed.

Thermal and chemical pollution:

- Coal is washed with water from lakes and rivers.
- Dirty water goes back but is now polluted with chemicals.
- Power plant equipment is cooled with river and lake water.
- Changes in temperature and chemicals disturb the natural balance of the environment.

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Circuit symbols	Ohm's Law
Series and parallel circuits	Calculating electricity
Ammeters and voltmeters	Resistors
Circuits in texts, sketches and diagrams	Movement and electricity
Predicting problems in changed electrical circuits	Electricity from coal

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